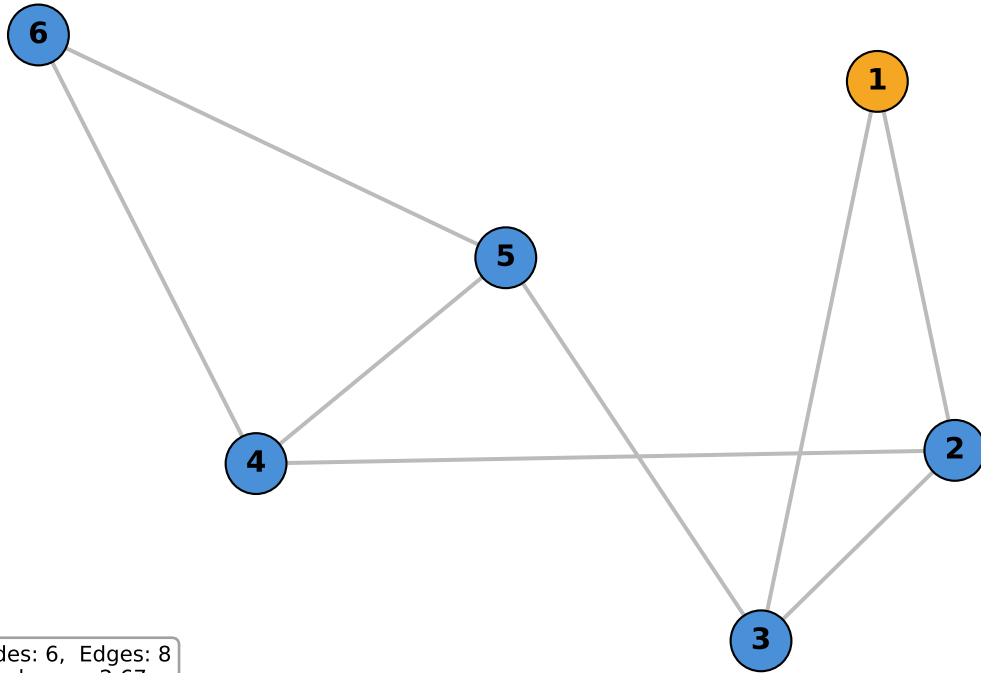


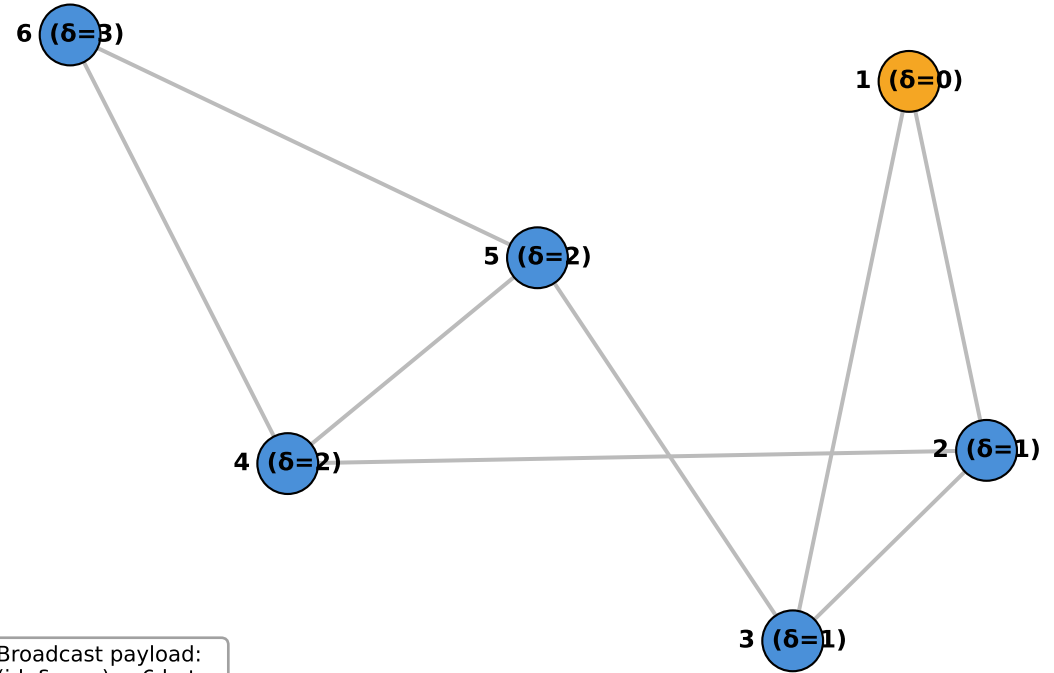
Comm-Feasible δ -BFS Spanning-Tree Pruning — Phase Overview

Phase 1: Initial communication graph $\mathcal{G}$



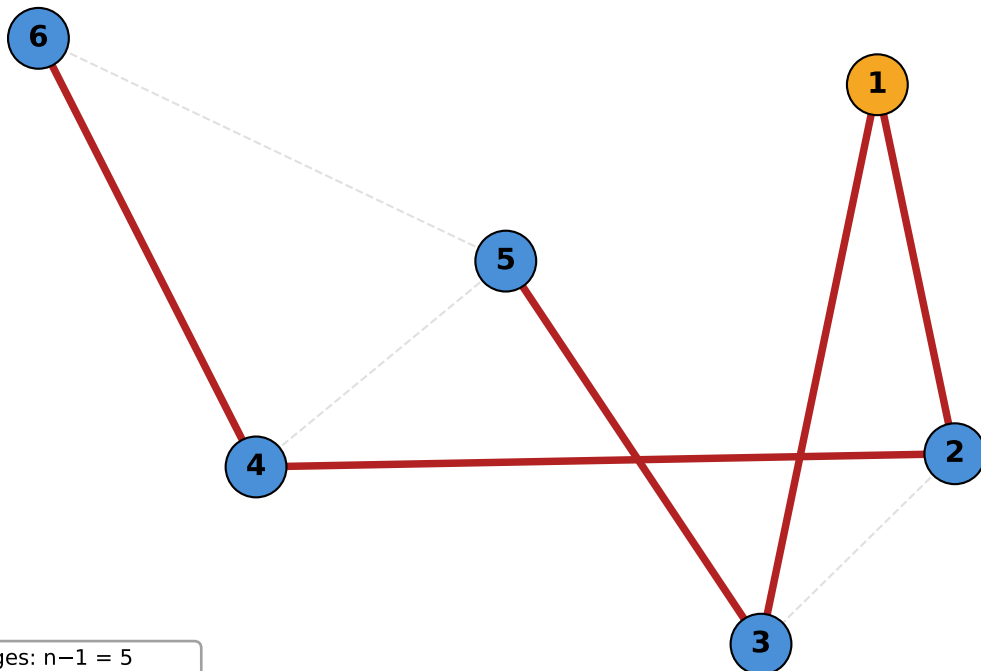
Nodes: 6, Edges: 8
Avg degree: 2.67

Phase 2: δ -BFS hop-to-root propagation
(constant-size messages)



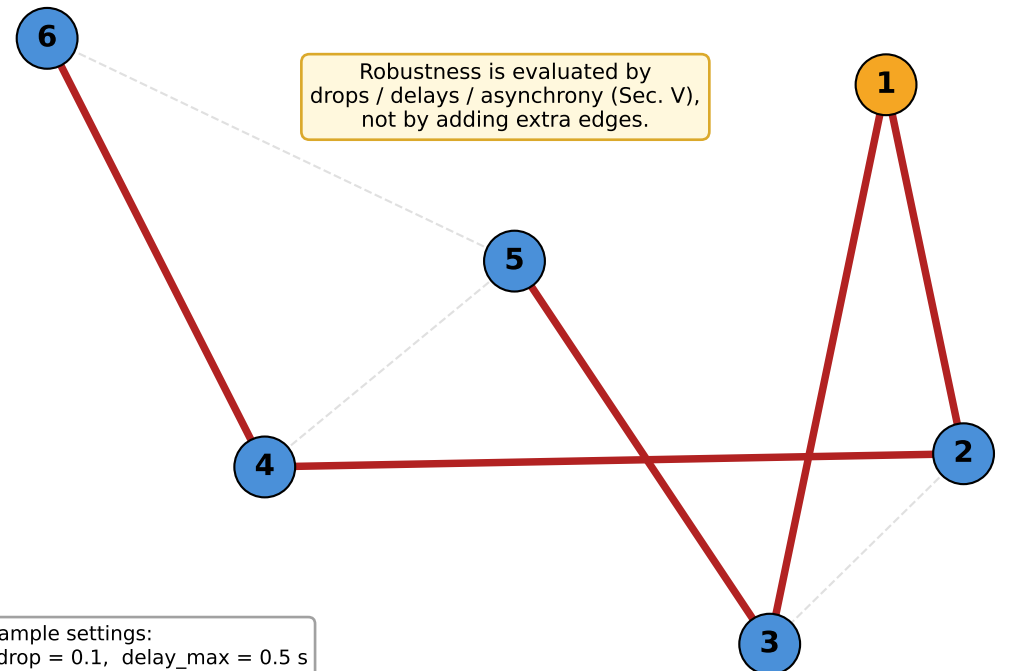
Broadcast payload:
(id, δ , seq) = 6 bytes

Phase 3: Pruned spanning tree $\mathcal{G}_{\text{topo}}$



Edges: $n-1 = 5$
✓ Tree ✓ Connected

Phase 4: Robustness evaluated via
comm impairments (no extra edges)



Robustness is evaluated by
drops / delays / asynchrony (Sec. V),
not by adding extra edges.

Example settings:
 $p_{\text{drop}} = 0.1$, $\text{delay}_{\text{max}} = 0.5$ s
 $\text{jitter} = 0.2$ s $\rightarrow$ tree formed ✓